



Matteo Ceradini

AI ENGINEER

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Summary

I'm an **AI & Software Engineer** with a PhD in Biorobotics and a background in Computer Science, combining expertise in **machine learning, scientific research, and production AI systems**. Currently developing and deploying LLM-powered agents for banking support and data analysis, **working with generative AI, LangGraph, RAG, and context engineering**. During my research career, I developed and evaluated deep learning models for complex biomedical signal-processing tasks, **combining machine learning with experimental research involving human participants**.

I am particularly interested in **applying AI to challenging real-world problems** and developing **reliable AI solutions** for **high-impact scientific and healthcare applications**.

Education

PhD in Biorobotics
Sant'Anna School of Advanced Studies, 2021 - 2025

Master in Computer Science (Artificial Intelligence course)
University of Torino, 2018-2021

Grade: 110/110 with honors and distinction.

Bachelor in Computer Science
University of Torino, 2015-2018

Grade: 103/110.

Tech stack

Python (advanced)	Github (advanced)	Tensorflow (basic)
PyTorch (advanced)	Matlab (advanced)	MongoDB (basic)
LangGraph (advanced)	Docker (medium)	NodeJS (basic)
LangChain (advanced)	C# (medium)	ReactJS (basic)
Langfuse (advanced)	Unity 3D (medium)	R (basic)
SQL (advanced)		C/C++ (basic)
Java (advanced)		

Languages

Italian - Native

English - fluent (C1 level)

Experiences

AI & Software Engineer
ION Group, Dec 2025 - current

- Develop and deploy LLM-powered AI agents for banking support and data analysis, integrating RAG, knowledge graphs, and context engineering to improve the quality and reliability of AI-generated responses.
- Developed an agent-based solution for assessing legal signatory authority, combining LLMs with structured information retrieved from graph databases.
- Contribute to the development of software solutions for credit management, working across AI and traditional software engineering workflows.

PhD researcher in Neural engineering
Sant'Anna School of Advanced Studies, Oct 2021 - Sep 2025

Goal of my research was the innovation of decoding methods for neuroprosthetic applications in people with spinal cord injury. I developed deep learning methods for neural signal processing and decoding, including online decoding of movement intentions from brain or muscle signals. Designed and developed immersive VR-based experimental protocols to evaluate human-machine interaction and brain-machine interface control strategies in clinical settings for real-life target applications.

Visiting PhD in Biomedical Engineering
University of Michigan, Jul 2024 - Feb 2025

Visiting period in the Chestek Lab, working on the development of online BCI exploiting the usage of implantable neural signals. Main activities included online decoding of individual fingers movements from implanted EMG signal in non-human participants.

AI engineer
Cynexo & SISSA, May 2021 - Sep 2021

In collaboration with the Time Perception Lab at SISSA, I developed a proof-of-concept closed-loop system integrating deep learning models for real-time EEG signal decoding to control transcranial stimulation.

Experiences

AI engineer

Cynexo & SISSA, Sep 2020 - Mar 2021

In collaboration with the Visual Neuroscience Lab at SISSA research center I worked on the porting of the DeepLabCut neural network for animal pose estimation to an embedded platform. I implemented a knowledge distillation method that resulted in inference speed on the embedded device of ~10fps with an accuracy with an accuracy close to 100% respect to the original model.

AI Engineer intern

Zextras, Jul 2018 - Ago 2018

Two-month internship at Zextras during the completion of my bachelor's degree. Conducted research on the effectiveness of spam filtering over several thousand emails using deep learning techniques, and implemented a prototype achieving over 98% accuracy with a very low false positive rate.

Web developer

Tecnobit, Jun 2016 - Sep 2020

From 2016 to 2020, while completing my bachelor's and master's degrees, I worked part-time as a full-stack web developer. My responsibilities included both front-end and back-end tasks, using php frameworks such as Codeigniter and Laravel. During this time, I maintained and developed websites such as sketchupitalia.it, corsigeometri.it, topgeometri.it, and others.

Summer internships

Margraf project & Industrie Metalpress, Jun-Sep 2013 & Jun-Sep 2014

Summer internships at Margraf Project and Industrie Metalpress (2013-2014). At Margraf, I worked as a marble operator on polishing and finishing tasks. At Metalpress, I worked on company Intranet web-app and on developing a C# application for network security activity monitoring via SNMP protocol.

Certifications

Retrieval Augmented Generation (RAG)

DeepLearning.ai, Sep 2025

End-to-end RAG system design and implementation; retrieval strategies, knowledge base integration, and optimization trade-offs (cost, latency, quality)

Fundamentals of Reinforcement Learning

University of Alberta, Alberta Machine Intelligence Institute by Coursera, Oct 2021

Markov Decision Processes, exploration-exploitation trade-offs, value function methods, and dynamic programming for control and optimization

Awards & Fellowships

Best paper presented by a young researcher

IEEE MetroXRAINE 2023 Conference

This award recognizes the most outstanding paper presented by a Young Researcher at IEEE MetroXRAINE 2023.

Zegna Scholarship Fellowship

IEEE MetroXRAINE 2023 Conference

Recipient of the Zegna Scholarship, funding my 8-months research stay at the University of Michigan where I worked on Brain and Body Machine Interfaces using implantable signals for neuroprosthetics applications.

Publications on international journals

Easy-to-use EMG system to decode hand movements in people with spinal cord injury

Ceradini et. al. 2026, Journal of Neural Engineering

The effect of User Learning of Online EEG Decoding of Upper-Limb Movement Intention

Ceradini et. al. 2025, IEEE Transactions on Medical Robotics and Bionics

A Virtual Reality-Based Protocol to Determine the Preferred Control Strategy for Hand Neuroprostheses in People With Paralysis

Losanno*, Ceradini* et. al. 2024 (* equally contributing), IEEE Transactions on Neural Systems and Rehabilitation Engineering

Immersive VR for Upper-Extremity Rehabilitation in Patients with Neurological Disorders: a scoping review

Ceradini et. al. 2024, Journal of NeuroEngineering and Rehabilitation